

SATHWIK KARRA

Phone: 07442786496

Email: karsath2026@gmail.com

SUMMARY

Data Engineer with experience in building and optimising data pipelines using Azure Data Factory, Databricks, and ADLS Gen2.Specialised in Medallion Architecture, PySpark transformations, and data governance via Unity Catalog. Delivered 25% reduction in Power BI report load times while processing 2TB/month for an enterprise client.

SKILLS

Languages:	Python, SQL
Big Data:	Apache Spark, PySpark
Data Warehouse:	Snowflake
Visualization:	Power BI
Version Control:	Git, GitHub
Cloud & Storage:	Azure ADLS, Databricks, Azure Synapse Analytics, Snowflake
Orchestration & Architecture:	Azure Data Factory
Data Modeling:	Star Schema & Snowflake
Project Management:	Agile, JIRA
File Formats:	Delta Lake, Parquet, Avro, JSON

PROFESSIONAL EXPERIENCE

Client: Amey (Spa plc)	July 2025 – Till date
Data Engineer	
Responsibilities:	- Engineered and optimized batch ETL pipeline in Azure Data Factory and Databricks, ingesting 2TB/month from e-commerce APIs and into Azure Data Lake, significantly improving pipeline reliability and data availability. - Architected modular data transformation using Medallion Architecture (Bronze, Silver, Gold) in Databricks with PySpark, Python, and SQL, increasing processing efficiency. - Designed a star schema data model with SCD Types 1 and 2, reducing Power BI reports load time by 25% for business users. - Enforced data governance using Unity Catalog, including role-based access control and data lineage, granted controlled access to the Silver layer table for downstream teams, improving data quality and consistency for ML workflows. - Standardised with Git based version Control workflows for Azure Data Factory pipelines, reducing manual deployment errors and accelerating team delivery efficiency - Collaborates with cross-functional teams to design and implement a scalable ETL pipeline and provide data for reporting and analytics. - Delivered ETL pipeline features across multiple Agile sprints, collaborating with data analysts and ML engineers to meet business reporting requirements on time

Data Engineer

Responsibilities:

- Built a batch **ETL** pipeline in Snowflake to ingest raw insurance claims and policy data from AWS S3, Configuring external stages and file formats for structured data loading.
- Implemented modular data transformation using raw layer, staged layer, and presentation layer.
- Automated data ingestion using Snowpipe, so files landing in S3 were picked up and loaded into Snowflake instantly without any manual trigger.
- Designed the **COPY** command for bulk loads and set up the staging table to hold the raw data.
- Handled data quality issues, including duplicates, nulls, type mismatches, and derived columns, before data moved to the next layer.
- Set up a reject table to capture failed quality checks, making it straightforward to investigate and fix bad data.
- Modelled insurance claims and policy data using a star schema with SCD Type2 to support accurate historical reporting in Power BI and reduced loading time by 20 %.
- Configured Snowflake tasks for batch scheduling and used Time Travel for data recovery when production issues arose.

Database Engineer (Internship)

Edinburgh Napier University

Jan 2023-May 2023

- Designed, developed, and maintained SQL database to meet the requirements, ensuring data integrity and Performance.
- Created and managed database tables, including data insertion, updates, modifications, and retrieval using SQL queries (SELECT, JOINS, GROUPBY, etc..)
- Developed optimized SQL scripts for data extraction and transformation to support business Intelligence needs.
- Gained expertise in SSMS, MySQL, and phpMyAdmin, enhancing database management capabilities.
- Worked with a version control system managing repositories, branches, commits, pull requests, and merges for seamless collaboration.

EDUCATION

Master's in Computer Science

Edinburgh Napier University

Jan 2022- Jul 2024

- Sentiment analysis in Software Engineering: Worked with sentiment analysis tools such as SentiCR, Senti4SD, DEVA, SentiStrength, and Savioka BERT to analyze developer sentiments in the software engineering context.
- Programming & Scripting: Proficient in Python, specializing in writing efficient scripts and implementing data structures for optimized performance.
- Data Processing & Analysis: Skilled in data cleaning techniques using pandas and Numpy.